



Chapter 1: Introduction to Derivatives Markets (Continued)

10. Dangers of Derivatives (Section 1.10)

a. Versatility and Misuse

- **Problem:** The high versatility of derivatives (used for hedging, speculation, and arbitrage) is also a source of great danger.
- **Rogue Traders:** Traders whose mandate is to hedge or arbitrage can become **speculators** (consciously or unconsciously), leading to disastrous losses.

b. Examples of Rogue Trader Losses

- **Jérôme Kerviel (Société Générale, 2008):**
 - **Role:** Junior trader mandated to find **arbitrage opportunities** (e.g., inconsistencies between futures and cash markets).
 - **Action:** Used knowledge of bank procedures to enter huge, **unauthorized, unhedged speculative positions** in equity indices, while creating **fictitious trades** to make it appear he was hedged.
 - **Outcome:** When uncovered, the position resulted in a loss of **4.9 billion euros**, at the time the biggest loss from fraudulent activity in finance history.
- **Nick Leeson (Barings Bank, 1990s):**
 - **Role:** Mandated to arbitrage Nikkei 225 futures.
 - **Action:** Made large speculative bets on the index direction, losing **\$1 billion** and leading to the **destruction of the 200-year-old bank**.
- **John Rusnak (Allied Irish Bank, 2002):** Lost **\$700 million** from unauthorized foreign exchange trading.

c. Corporate Controls and Lessons Learned

- **Internal Controls:** Financial and nonfinancial corporations must implement controls to ensure derivatives are used only for their **intended purpose**.
- **Risk Management:**
 - **Set Risk Limits:** Unambiguous risk limits must be clearly defined for traders.
 - **Monitor Activity:** Trader activities must be monitored daily to ensure adherence to limits.

- **The Credit Crisis Lesson:** The period leading up to the 2007 credit crisis showed that even following specified risk limits can be insufficient if the underlying **risks are fundamentally misjudged**.
 - **Mistake:** Financial institutions failed to prepare for the **steep decline in US house prices** and the **high correlation between mortgage defaults** across different regions.
 - **Key Lesson:** Financial institutions should **always dispassionately ask**:
 - a. "What can go wrong?"
 - b. "If it does go wrong, how much will we lose?"

11. Chapter Summary and Unifying Framework (Section 1.10 Summary)

a. Derivatives Fundamentals

- **Success:** The growth of derivatives markets has been one of the most exciting developments in finance over the last 40 years.
- **Attraction:** Both hedgers and speculators often find it more attractive to trade a **derivative on an asset** than to trade the asset itself.
- **Markets:** Derivatives trade on both **Exchanges** (standardized contracts) and in the **Over-the-Counter (OTC) market** (customized contracts).
- **Valuation Aim:** A major aim of studying derivatives is to present a **unifying framework** for the valuation of *all* derivatives.

b. Contract Types

Contract Type	Holder's Obligation/Right	Key Feature
Forward or Futures	Obligation to buy or sell an asset at a future time for a certain price.	Creates a binding commitment.
Call Option	Right to buy an asset by a certain date for a certain price.	Provides potential upside benefit.
Put Option	Right to sell an asset by a certain date for a certain price.	Provides protection against downside risk.

c. Trader Categories Revisited

Trader Type	Action and Motivation	Key Tool Feature Used
Hedgers	Reduce or eliminate risk associated with asset prices.	Neutralizes or insures against price movements.
Speculators	Bet on future price movements of an asset.	Provides leverage (large position with small outlay).
Arbitrageurs	Take advantage of discrepancies in prices between two markets to lock in a riskless profit.	Enforces the no-arbitrage principle in pricing.